WORKED SOLUTION

STEP I 1998, Question 12

Probability and Statistics • Counting and Independence

STEP I 1998, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s1-q12

The key idea

Everything here is counting. Once we know how many solutions there are, and how many have X equal to a given value, each probability is a ratio of two counts. For independence, the quickest test is an extreme case: a single pair of values where the joint probability can't possibly equal the product of the marginal probabilities. For the last part, "XYZ divisible by 5" means "at least one of X, Y, Z is divisible by 5", which calls for inclusion–exclusion.

Solution

Counting the solutions

If $X = x$ is fixed (with $0 \le x \le 20$), then Y can be any of $0, 1, \ldots, 20 - x$, and $Z = 20 - x - Y$ is then determined and non-negative. So exactly $21 - x$ solutions have $X = x$, and the total number of solutions is

$$N = \sum_{x=0}^{20} (21 - x) = 21 + 20 + \cdots + 1 = \frac{21 \cdot 22}{2} = 231.$$

(This is the "stars and bars" count $\binom{22}{2}$: arrange 20 stars and 2 bars in a row.) Each solution has probability $1/231$, and

$$\mathrm{P}(X = x) = \frac{21 - x}{231}, \qquad x = 0, 1, \ldots, 20.$$

By symmetry, Y (and Z) has the same distribution.

Are X and Y independent?

No. Take $X = 20$ and $Y = 20$. Each event is possible on its own:

$$\mathrm{P}(X = 20) = \mathrm{P}(Y = 20) = \frac{1}{231} > 0,$$

but they can't happen together, since $X + Y \leq 20$. So

$$\mathrm{P}(X = 20,\ Y = 20) = 0 \neq \frac{1}{231^2} = \mathrm{P}(X = 20)\,\mathrm{P}(Y = 20),$$

and X and Y are not independent. Intuitively, a large value of X leaves less room for Y.

> **Ask yourself** Isn't it enough to say "$X + Y \leq 20$, so they're obviously dependent"?

A constraint like that is the right instinct, but independence is a precise statement about probabilities, so the justification should exhibit one pair of values where $\mathrm{P}(X = x, Y = y) \neq \mathrm{P}(X = x)\mathrm{P}(Y = y)$. A single counterexample is all it takes, and the extreme pair $(20, 20)$ makes the arithmetic trivial.

X divisible by 5

The multiples of 5 in range are $X = 0, 5, 10, 15, 20$, with $21, 16, 11, 6, 1$ solutions respectively. So

$$\mathrm{P}(5 \mid X) = \frac{21 + 16 + 11 + 6 + 1}{231} = \frac{55}{231} = \frac{5}{21},$$

as required.

XYZ divisible by 5

Since 5 is prime, $5 \mid XYZ$ exactly when 5 divides at least one of X, Y, Z. (Remember that 0 counts as divisible by 5.) Let A, B, C be the sets of solutions in which X, Y, Z respectively are divisible by 5. From the previous part, $|A| = |B| = |C| = 55$.

For $A \cap B$: write $X = 5a$, $Y = 5b$ with $a, b \geq 0$ and $a + b \leq 4$. The number of such pairs is $5 + 4 + 3 + 2 + 1 = 15$. In every one of them $Z = 20 - 5a - 5b$ is also a multiple of 5, so

$$|A \cap B| = |B \cap C| = |C \cap A| = |A \cap B \cap C| = 15.$$

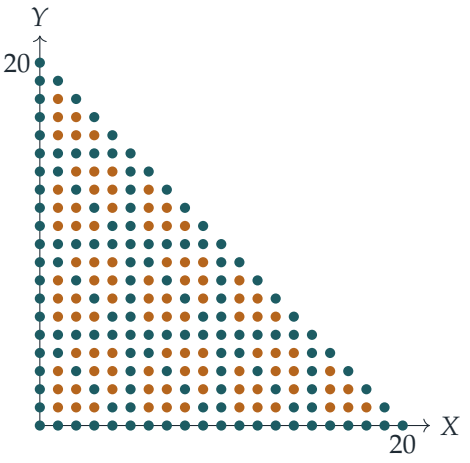
By inclusion–exclusion,

$$\begin{aligned} |A \cup B \cup C| &= 3 \cdot 55 - 3 \cdot 15 + 15 \\ &= 135, \end{aligned}$$

and therefore

$$\mathrm{P}(5 \mid XYZ) = \frac{135}{231} = \frac{45}{77}.$$

The picture shows the solutions as lattice points (X, Y) with $X + Y \leq 20$. The teal points lie on one of the lines $X = 5k$, $Y = 5k$ or $X + Y = 20 - 5k$ (that is, $Z = 5k$), and the amber points are the 96 solutions where none of X, Y, Z is a multiple of 5.



Checks

- The complement should have $231 - 135 = 96$ solutions in which none of X, Y, Z is a multiple of 5: the amber points in the picture. Listing all 231 solutions by computer, with exact fractions, confirms 96, $|A \cap B| = 15$, $5/21$ and $135/231 = 45/77 \approx 0.584$.
- A sanity check on $5/21$: X divisible by 5 is a bit more likely than $1/5$, because small values of X are the most likely ones and $X = 0$ is the most likely of all.

Common slips

- Forgetting that 0 is divisible by 5. Leaving out $X = 0$ gives $34/231$, not $5/21$.
- Adding $3 \times \frac{5}{21}$, which counts the overlaps several times. Here the overlaps are large: whenever two of the variables are multiples of 5, so is the third.
- Arguing for dependence without giving a specific pair of values where the product rule fails.

Beyond STEP

Choosing a solution uniformly at random is choosing a random *composition* of 20 into three parts. If 20 is replaced by a total n and n grows, $(X/n, Y/n)$ becomes uniformly distributed on a triangle, the simplest case of the Dirichlet distribution. Inclusion–exclusion is the standard tool for "at least one" events, and it appears again in derangements and in Euler's totient function. Grimmett and Welsh, *Probability: An Introduction*, covers both the counting and the independence ideas.

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